

Healthcare Analytics

Doctor Visits Analysis — Project Documentation

Python · MySQL · Power BI

5,190 Patients · 79.8% Zero Visits · 0.302 Avg Visits · Australia 1977-78

TOOLS	SOURCE	RECORDS	SCOPE
Python · MySQL · Power BI	Australian Health Survey	5,190 · 14 Cols	1977-78 National Survey

01 EXECUTIVE SUMMARY

- **Analyzed 5,190 patient records** from the classic Australian Health Survey (1977-78) doctor-visit dataset — genuinely clean at the source, with zero missing values and zero duplicates, requiring interpretive rather than corrective cleaning: un-scaling age and income fields that were stored as compressed ratios (age ÷ 100, income ÷ \$10,000 AUD).
- **Answered 9 structured health-behavior questions** covering visit frequency, gender, insurance status, chronic conditions, and 4 separate correlation analyses — then reproduced every answer independently in MySQL, including 4 hand-built Pearson correlation formulas.
- **Built a 2-page, 11-visual interactive Power BI dashboard** with 5 DAX measures, a zero-inflation distribution chart, and two scatter plots isolating the strongest predictors of visit frequency — connected directly to MySQL.
- **Surfaced a genuinely counter-intuitive finding:** having private health insurance shows essentially no relationship with visit frequency (0.295 vs. 0.307 average visits) — while self-reported illness count ($r = 0.224$) and a limiting chronic condition (0.603 vs. 0.262 average visits, a 2.3x gap) are the two strongest drivers found in the entire dataset.

02 PROBLEM STATEMENT

The dataset captures self-reported doctor-visit behavior from a large-scale 1977-78 Australian health survey, with each row representing one respondent's visits, demographics, insurance status, and health indicators over a 2-week reference window. Because two of the numeric fields (age, income) were stored in compressed, non-obvious scales, and because the target variable (visits) is heavily zero-inflated, a naive analysis risks either misinterpreting the demographic fields or drawing conclusions from a distribution shape that isn't well summarized by a simple mean. The goal of this project was to correctly interpret the data's scaling conventions, answer 9 concrete questions about who visits doctors and why, and validate every finding across Python, SQL, and an interactive dashboard.

03 OBJECTIVES

- **Obj 1:** Correctly identify and document the dataset's scaling conventions (age, income) before drawing any conclusions from them.
- **Obj 2:** Characterize the zero-inflated visits distribution rather than defaulting to a mean-only summary that would misrepresent it.

- **Obj 3:** Test 4 candidate predictors of visit frequency (age, income, illness, health score) and rank them by correlation strength.
- **Obj 4:** Reproduce all 9 answers independently in MySQL, including correlation formulas MySQL doesn't provide natively.
- **Obj 5:** Build a Power BI dashboard that visually foregrounds the strongest predictors rather than treating all variables as equally important.
- **Obj 6:** Produce a portfolio-ready deliverable demonstrating careful interpretation of a deceptively simple, already-clean dataset.

04 DATASET OVERVIEW

Source & Structure

Attribute	Detail
Dataset Name	DoctorVisits (Australian Health Survey, 1977-78)
File Format	CSV, single table
Records	5,190 rows × 13 columns raw → 14 columns clean (2 derived columns added, 1 index column dropped)
Missing Values	Zero across all columns
Duplicate Rows	Zero
Age Scaling	Stored as age ÷ 100 — actual range 19–72 years
Income Scaling	Stored as income ÷ \$10,000 AUD — actual range \$0–\$15,000 AUD (era-appropriate, not an error)
Target Variable Shape	visits is heavily zero-inflated: 79.8% of respondents had zero visits

Key Columns Used in Analysis

Column	Type	Description
visits	Integer	Doctor visits in the 2-week reference window, 0–9
gender	Text	female / male
age_years	Decimal	Un-scaled age in years, 19–72 (derived from age × 100)
income_aud	Decimal	Un-scaled annual income in AUD, \$0–\$15,000 (derived from income × 10,000)
illness	Integer	Self-reported illness count, past 2 weeks, top-coded at 5
health	Integer	General health questionnaire score, 0–12 — higher = worse health
private	Text	Has private health insurance: yes / no
lchronic	Text	Has a limiting chronic condition: yes / no
freerepat	Text	Free government insurance via age/disability/veteran status: yes / no

05 TOOLS & TECHNOLOGIES USED

Tool	Platform	Purpose in This Project
Python 3	VS Code	Data loading, interpretive cleaning, EDA, correlation analysis
pandas / numpy	Python library	Scale conversion, groupby aggregation, correlation computation
MySQL 8	MySQL Workbench	Relational storage; SQL validation of all 9 answers
Power BI Desktop	Microsoft	2-page dashboard, DAX measures, scatter analysis of top predictors
DAX	Power BI native language	5 measures: Total Patients, Avg Visits, % Zero Visits, Avg Illness Count, Avg Health Score

06 DATA CLEANING PROCESS — PYTHON

This dataset required almost no corrective cleaning — the real work was correct interpretation of its scaling conventions.

Step	Action	Result
1	Load CSV, confirm zero duplicates and zero missing values	5,190 rows × 13 columns confirmed clean
2	Drop the Unnamed: 0 redundant row-index column	13 → 12 columns
3	Derive age_years = age × 100 for readable interpretation	Readable 19–72 year range
4	Derive income_aud = income × 10,000 for readable interpretation	Readable \$0–\$15,000 AUD range
5	Confirm all 6 yes/no columns are clean 2-value categoricals	No fixing required — verified, not assumed

07 SQL / MYSQL — VALIDATION LAYER

A single flat table (doctor_visits), imported directly, with 3 indexes on gender, private, and lchronic. All 9 Python answers were reproduced independently in MySQL to 3 decimal places.

Four Correlations, Zero Native Support

MySQL has no CORR() aggregate. All 4 correlation questions (age, income, illness, health — each against visits) were implemented manually using the Pearson formula: $r = (n\sum xy - \sum x \sum y) / \sqrt{[(n\sum x^2 - (\sum x)^2)(n\sum y^2 - (\sum y)^2)]}$. All 4 results matched Python to 3+ decimal places: illness 0.2236, health 0.1933, age 0.1245, income -0.0768.

Analytical VIEW

A vw_healthcare_summary view buckets patients by age (4 bands) and illness count (3 bands) crossed with gender, private, freepoor, freerepat, nchronic, and lchronic — enabling a VIEW Weighted Avg Visits measure that cross-validates the row-level average.

08 DAX MEASURES CREATED

Measure	DAX Formula	Result
Total Patients	COUNTROWS(doctor_visits)	5,190

Measure	DAX Formula	Result
Avg Visits	AVERAGE(doctor_visits[visits])	0.302
% Zero Visits	DIVIDE(CALCULATE([Total Patients], visits=0), [Total Patients])*100	79.79
Avg Illness Count	AVERAGE(doctor_visits[illness])	~1.24
Avg Health Score	AVERAGE(doctor_visits[health])	~1.22

Formatting note: % Zero Visits already multiplies by 100 in DAX. Displaying it correctly requires a Custom number format (0.00"%") rather than Power BI's built-in Percentage type, which would double the multiplication (79.79 → 7,979%).

09 DASHBOARD VISUALS BUILT (11 Visuals, 2 Pages)

Page 1 — Visit Patterns Overview

#	Visual Type	Fields Used	Insight Delivered
1-3	KPI Cards	Total Patients / Avg Visits / % Zero Visits	5,190 patients, 0.302 avg visits, 79.8% had none
4	Column Chart	X: visits (category) / Y: Total Patients	The zero-inflation is immediately visible as the dominant bar
5	Clustered Column	X: gender / Y: Avg Visits	Female 0.362 vs. Male 0.236 — a clear, real gap
6	Clustered Column	X: private / Y: Avg Visits	0.295 vs 0.307 — visually confirms almost no difference
7-8	Slicers	gender, private	Cross-filters all Page 1 visuals

Page 2 — Health Drivers & Chronic Conditions

#	Visual Type	Fields Used	Insight Delivered
1	Scatter + trend line	X: illness / Y: visits (Don't summarize)	Strongest predictor found, $r = 0.224$
2	Scatter + trend line	X: health / Y: visits	Second-strongest, $r = 0.193$ (higher score = worse health)
3	Clustered Column	X: lchronic / Y: Avg Visits	0.603 vs 0.262 — more than double for limiting chronic conditions
4	Clustered Column	X: freerepat / Y: Avg Visits	0.467 vs 0.258 — a secondary but real gap
5	Slicer	lchronic	Cross-filters Page 2 visuals

10 KEY INSIGHTS & RESULTS

A. Visit Distribution

- **The vast majority of respondents saw no doctor at all:** 79.8% (4,141 of 5,190) had zero visits in the 2-week window; only 20.2% had any contact at all, and the max was 9 visits.

B. Demographic Predictors

- **Gender is a real, moderate predictor:** women average 0.362 visits versus men's 0.236 — roughly 53% more frequent.
- **Age and income are both weak predictors:** age correlates at only $r = 0.125$ and income at $r = -0.077$ — neither is a meaningful driver on its own.

C. The Insurance Surprise

- **Private insurance shows essentially no relationship with visit frequency:** 0.295 average visits for insured patients versus 0.307 for uninsured — the opposite direction from what's often assumed, and small enough to be noise rather than signal.

D. The Two Strongest Drivers

- **Self-reported illness count is the single strongest predictor found:** $r = 0.224$ with visits — stronger than gender, age, income, or insurance status combined.
- **A limiting chronic condition more than doubles average visits:** 0.603 for patients with `lchronic='yes'` versus 0.262 for those without — the largest gap of any binary variable tested, and the general health questionnaire score ($r = 0.193$) reinforces the same underlying signal.

11 CHALLENGES FACED & HOW OVERCOME

Challenge 1 — Deceptive Scaling on Two Key Fields

Problem: age and income were stored as compressed ratios ($\div 100$ and $\div 10,000$ respectively) with no in-file documentation — easy to misinterpret as raw values and draw wrong conclusions from (e.g. treating `age=0.45` as a 45-year age gap rather than 45 years old).

Solution: Cross-referenced the value ranges (0.19–0.72 for age) against known conventions for this specific, well-documented survey dataset, then created explicit `age_years` and `income_aud` derived columns so every downstream chart and table would be unambiguous.

Challenge 2 — Four Correlations, No Native SQL Support

Problem: MySQL has no `CORR()` function, and this project needed 4 separate correlation checks, not just one.

Solution: Built the Pearson formula once as a reusable SQL pattern, then swapped in each of the 4 predictor columns — validated all 4 against Python to 3+ decimal places rather than trusting the formula's correctness on the first pair alone.

Challenge 3 — A Ratio Measure Displaying Without Its Percent Sign

Problem: % Zero Visits initially rendered as a bare "79.79" with no percent symbol, since the DAX formula already multiplies by 100 and Power BI's Percentage format would have doubled that.

Solution: Applied a Custom number format (0.00"%"") that appends the symbol as a literal character without re-triggering the multiplication — a distinct fix from the more common "forgot to multiply" version of this bug seen in other projects.

12 SKILLS LEARNED & TAKEAWAYS

- Recognizing when a dataset needs interpretive rather than corrective cleaning — zero missing values and zero duplicates doesn't mean zero cleaning work.
- Correctly un-scaling compressed numeric fields by reasoning from value ranges rather than assuming units.

- Characterizing a zero-inflated distribution properly instead of defaulting to a mean that would misrepresent 80% of the data.
- Building one reusable Pearson-correlation SQL pattern and validating it across multiple variable pairs, rather than treating each correlation as a one-off.
- DAX percentage-formatting nuance: distinguishing the "forgot to multiply" bug from the "multiplied correctly but formatted wrong" bug — they look identical in the DAX code but need opposite fixes.
- Letting counter-intuitive findings (insurance status barely mattering) stand as reported results rather than searching for a confound to explain them away.

13 CONCLUSION

- **Analyzed 5,190 patient records** from a genuinely clean source dataset, correctly interpreting two compressed-scale fields rather than treating "zero missing values" as "zero work required."
- **Independently validated all 9 project questions in MySQL**, including 4 separate hand-built Pearson correlation formulas, confirming agreement with Python to 3+ decimal places.
- **Delivered a 2-page, 11-visual Power BI dashboard** with 5 DAX measures correctly formatted to avoid a percentage-display trap, foregrounding illness count and chronic-condition status as the two strongest visit-frequency predictors.
- **Most notably, let the data override intuition:** private insurance status — often assumed to drive healthcare utilization — showed essentially no relationship with visit frequency, while self-reported illness and chronic-condition status did. Reporting that null result accurately was as important as reporting the real ones.

KPI Summary

Metric	Value	Metric	Value
Total Patients	5,190	Zero-Visit Share	79.8%
Avg Visits	0.302	Max Visits Recorded	9
Female Avg Visits	0.362	Male Avg Visits	0.236
Private Insurance Avg	0.295	No Insurance Avg	0.307
Strongest Predictor	Illness Count ($r=0.224$)	2nd Strongest	Health Score ($r=0.193$)
Limiting Chronic Avg	0.603	No Chronic Condition Avg	0.262
Age Range	19–72 years	Income Range	\$0–\$15,000 AUD
Missing Values	0	Duplicate Rows	0